

MODULE 4

Biomass Energy, Bio Gas, Tidal Energy

Biomass is biological material from living, or recently living organisms, most often referring to plants or plant-derived materials. As a renewable energy source, biomass can either be used directly, or indirectly -- once or converted into another type of energy product such as biofuel. Biomass can be converted to energy in three ways: thermal conversion, chemical conversion, and biochemical conversion.

Historically, humans have harnessed biomass derived energy products since the time when people began burning wood to make fire. In modern times, the term can be referred to in two meanings. In the first sense, biomass is plant matter used either to generate electricity (via steam turbines or gasifiers), or to produce heat (via direct combustion). Wood remains the largest biomass energy source today; examples include forest residues (such as dead trees, branches and tree stumps), yard clippings, wood chips and even municipal solid waste. In the second sense, biomass includes plant or animal matter that can be converted into fibers or other industrial chemicals, including biofuels. Industrial biomass can be grown from numerous types of plants, including miscanthus, switchgrass, hemp, corn, poplar, willow, sorghum, sugarcane, bamboo, and a variety of tree species, ranging from eucalyptus to oil palm (palm oil).

The adoption of biomass-based energy plants has been a slow but steady process. Over the past decade, the production of these plants has increased 14%. In the United States, alternative electricity-production sources on the whole generate about 13% of power; of this fraction, biomass contributes approximately 11% of the alternative production. According to a study conducted in early 2012, of the 107 operating biomass plants in the United States, 85 have been cited by federal or state regulators for the violation of clean air or water standards laws over the past 5 years. This data also includes minor infractions.

Biomass mass derived energy also holds the promise of reducing carbon dioxide emissions, a significant contributor to global warming, carbon dioxide acts as a “greenhouse” gas by trapping heat absorbed by the earth from the sun. Although the burning of biomass energy releases as much carbon dioxide as fossil fuels, biomass burning does not release “new carbon” into the atmosphere while burning fossil fuels does. This is because carbon dioxide released from fossil fuels was carbon that was fixated via photosynthesis millions of years ago that had been locked in the hydrocarbons of fossil fuels.

Industry professionals claim that a range of issues can affect a plant's ability to comply with emissions standards. Some of these challenges, unique to biomass plants, include inconsistent fuel supplies and age. The type and amount of the fuel supply is completely reliant factors; the fuel can be in the form of building debris or agricultural waste (such as deforestation of invasive species or orchard trimmings). Furthermore, many of the biomass plants are old, use outdated technology and were not built to comply with today's stringent standards. In fact, many are based on technologies developed during the term of President Jimmy Carter, who created the Department of Energy in 1977.

The Energy Information Administration projected that by 2017, biomass is expected to be about twice as expensive as natural gas, slightly more expensive than nuclear power, and much less expensive than solar panels. In another EIA study released, concerning the government's plan to implement a 25% renewable energy standard by 2025, the agency assumed that 598 million tons of biomass would be available, accounting for 12% of the renewable energy in the plan

PRINCIPLES FOR PRODUCTION OF BIOGAS:

Organic substances exist in wide variety from living beings to dead organisms. Organic matters are composed of Carbon (C), combined with elements such as Hydrogen (H), Oxygen (O), Nitrogen (N), Sulphur (S) to form variety of organic compounds such as carbohydrates, proteins & lipids. In nature MOs (microorganisms), through digestion process breaks the complex carbon into smaller substances.

There are 2 types of digestion process:

Aerobic digestion.

Anaerobic digestion.

The digestion process occurring in presence of Oxygen is called Aerobic digestion and produces mixtures of gases having carbon dioxide (CO₂), one of the main "green-houses" responsible for global warming.

The digestion process occurring without (absence) oxygen is called anaerobic digestion which generates mixtures of gases. The gas produced which is mainly methane produces 5200-5800 KJ/m³ which when burned at normal room temperature and presents a viable environmentally friendly energy source to replace fossil fuels (non-renewable

OBJECTIVES:

Optimization of gas production

Comparison with conventional plants

Effect of different parameters viz.

* Temperature

* PH

* Total & volatile solid concentration

* Alkalinity

* C:N Ratio

To increase the production by using * Additives

* Nutrients

* Nitrogen source

Photosynthesis:

Photosynthesis is the process of converting light energy to chemical energy and storing it in the bonds of sugar. This process occurs in plants and some algae (Kingdom Protista). Plants need only light energy, CO₂, and H₂O to make sugar. The process of photosynthesis takes place in the chloroplasts, specifically using chlorophyll, the green pigment involved in photosynthesis.

Photosynthesis takes place primarily in plant leaves, and little to none occurs in stems, etc. The parts of a typical leaf include the upper and lower epidermis, the mesophyll, the vascular bundle(s) (veins), and the stomates. The upper and lower epidermal cells do not have chloroplasts, thus photosynthesis does not occur there. They serve primarily as protection for the rest of the leaf. The stomates are holes which occur primarily in the lower epidermis and are for air exchange: they let CO₂ in and O₂ out. The vascular bundles or veins in a leaf are part of the plant's transportation system, moving water and nutrients around the plant as needed. The mesophyll cells have chloroplasts and this is where photosynthesis occurs.

As you hopefully recall, the parts of a chloroplast include the outer and inner membranes, intermembrane space, stroma, and thylakoids stacked in grana. The chlorophyll is built into the membranes of the thylakoids.

Chlorophyll looks green because it absorbs red and blue light, making these colors unavailable to be seen by our eyes. It is the green light which is NOT absorbed that finally reaches our eyes, making

chlorophyll appear green. However, it is the energy from the red and blue light that are absorbed that is, thereby, able to be used to do photosynthesis. The green light we can see is not/cannot be absorbed by the plant, and thus cannot be used to do photosynthesis.

The overall chemical reaction involved in photosynthesis is: $6\text{CO}_2 + 6\text{H}_2\text{O} (+ \text{light energy}) \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$. This is the source of the O_2 we breathe, and thus, a significant factor in the concerns about deforestation.

Stages of Photosynthesis:

Photosynthesis is a two stage process. The first process is the Light Dependent Process (Light Reactions), requires the direct energy of light to make energy carrier molecules that are used in the second process. The Light Independent Process (or Dark Reactions) occurs when the products of the Light Reaction are used to form C-C covalent bonds of carbohydrates. The Dark Reactions can usually occur in the dark, if the energy carriers from the light process are present. Recent evidence suggests that a major enzyme of the Dark Reaction is indirectly stimulated by light, thus the term Dark Reaction is somewhat of a misnomer. The Light Reactions occur in the grana and the Dark Reactions take place in the stroma of the chloroplasts.

Biofuel:

A biofuel is a type of fuel whose energy is derived from biological carbon fixation. Biofuels include fuels derived from biomass conversion, as well as solid biomass, liquid fuels and various biogases. Biofuels are gaining increased public and scientific attention, driven by factors such as oil price hikes and the need for increased energy security. However, according to the European Environment Agency, biofuels do not address global warming concerns.

Bioethanol is an alcohol made by fermentation, mostly from carbohydrates produced in sugar or starch crops such as corn or sugarcane. Cellulosic biomass, derived from non-food sources, such as trees and grasses, is also being developed as a feedstock for ethanol production. Ethanol can be used as a fuel for vehicles in its pure form, but it is usually used as a gasoline additive to increase octane and improve vehicle emissions. Bioethanol is widely used in the USA and in Brazil. Current plant design does not provide for converting the lignin portion of plant raw materials to fuel components by fermentation.

Biodiesel is made from vegetable oils and animal fats. Biodiesel can be used as a fuel for vehicles in its pure form, but it is usually used as a diesel additive to reduce levels of particulates, carbon

monoxide, and hydrocarbons from diesel-powered vehicles. Biodiesel is produced from oils or fats using transesterification and is the most common biofuel in Europe.

In 2010, worldwide biofuel production reached 105 billion liters (28 billion gallons US), up 17% from 2009, and biofuels provided 2.7% of the world's fuels for road transport, a contribution largely made up of ethanol and biodiesel.[citation needed] Global ethanol fuel production reached 86 billion liters (23 billion gallons US) in 2010, with the United States and Brazil as the world's top producers, accounting together for 90% of global production. The world's largest biodiesel producer is the European Union, accounting for 53% of all biodiesel production in 2010.[3] As of 2011, mandates for blending biofuels exist in 31 countries at the national level and in 29 states or provinces. According to the International Energy Agency, biofuels have the potential to meet more than a quarter of world demand for transportation fuels by 2050.

Bioalcohols:

Biologically produced alcohols, most commonly ethanol, and less commonly propanol and butanol, are produced by the action of microorganisms and enzymes through the fermentation of sugars or starches (easiest), or cellulose (which is more difficult). Biobutanol (also called bio gasoline) is often claimed to provide a direct replacement for gasoline, because it can be used directly in a gasoline engine (in a similar way to biodiesel in diesel engines).

Ethanol fuel is the most common biofuel worldwide, particularly in Brazil. Alcohol fuels are produced by fermentation of sugars derived from wheat, corn, sugar beets, sugar cane, molasses and any sugar or starch from which alcoholic beverages can be made (such as potato and fruit waste, etc.). The ethanol production methods used are enzyme digestion (to release sugars from stored starches), fermentation of the sugars, distillation and drying. The distillation process requires significant energy input for heat (often unsustainable natural gas fossil fuel, but cellulosic biomass such as bagasse, the waste left after sugar cane is pressed to extract its juice, can also be used more sustainably).

Ethanol can be used in petrol engines as a replacement for gasoline; it can be mixed with gasoline to any percentage. Most existing car petrol engines can run on blends of up to 15% bioethanol with petroleum/gasoline. Ethanol has a smaller energy density than that of gasoline; this means it takes more fuel (volume and mass) to produce the same amount of work. An advantage of ethanol ($\text{CH}_3\text{CH}_2\text{OH}$) is that it has a higher-octane rating than ethanol-free gasoline available at roadside gas stations, which allows an increase of an engine's compression ratio for increased thermal efficiency. In

high-altitude (thin air) locations, some states mandate a mix of gasoline and ethanol as a winter oxidizer to reduce atmospheric pollution emissions.

Ethanol is also used to fuel bioethanol fireplaces. As they do not require a chimney and are "flueless", bioethanol fires are extremely useful for newly built homes and apartments without a flue. The downside to these fireplaces is their heat output is slightly less than electric heat or gas fires.

In the current corn-to-ethanol production model in the United States, considering the total energy consumed by farm equipment, cultivation, planting, fertilizers, pesticides, herbicides, and fungicides made from petroleum, irrigation systems, harvesting, transport of feedstock to processing plants, fermentation, distillation, drying, transport to fuel terminals and retail pumps, and lower ethanol fuel energy content, the net energy content value added and delivered to consumers is very small. And, the net benefit (all things considered) does little to reduce imported oil and fossil fuels required to produce the ethanol.

Although corn-to-ethanol and other food stocks have implications both in terms of world food prices and limited, yet positive, energy yield (in terms of energy delivered to customer/fossil fuels used), the technology has led to the development of cellulosic ethanol. According to a joint research agenda conducted through the US Department of Energy, the fossil energy ratios (FER) for cellulosic ethanol, corn ethanol, and gasoline are 10.3, 1.36, and 0.81, respectively.

Even dry ethanol has roughly one-third lower energy content per unit of volume compared to gasoline, so larger (therefore heavier) fuel tanks are required to travel the same distance, or more fuel stops are required. With large current unsustainable, unscalable subsidies, ethanol fuel still costs much more per distance traveled than current high gasoline prices in the United States.

Methanol is currently produced from natural gas, a nonrenewable fossil fuel. It can also be produced from biomass as bio methanol. The methanol economy is an alternative to the hydrogen economy, compared to today's hydrogen production from natural gas.

Butanol (C_4H_9OH) is formed by ABE fermentation (acetone, butanol, ethanol) and experimental modifications of the process show potentially high net energy gains with butanol as the only liquid product. Butanol will produce more energy and allegedly can be burned "straight" in existing gasoline engines (without modification to the engine or car), and is less corrosive and less water-soluble than ethanol, and could be distributed via existing infrastructures. DuPont and BP are working together to help develop butanol. E. coli strains have also been successfully engineered to produce butanol by hijacking their amino acid metabolism.

Biogas:

Biogas is methane produced by the process of anaerobic digestion of organic material by anaerobes. It can be produced either from biodegradable waste materials or by the use of energy crops fed into anaerobic digesters to supplement gas yields. The solid byproduct, digestate, can be used as a biofuel or a fertilizer.

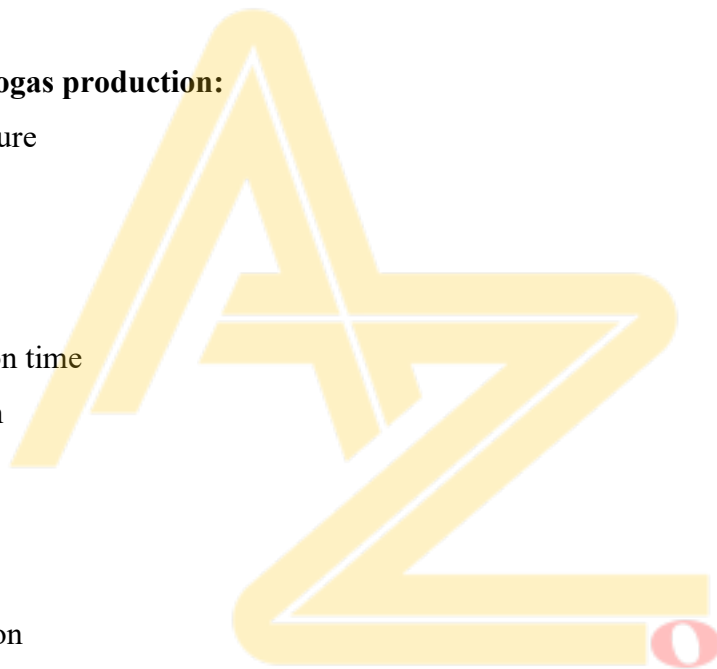
Biogas can be recovered from mechanical biological treatment waste processing systems.

Note: Land fill gas, a less clean form of biogas, is produced in landfills through naturally occurring anaerobic digestion. If it escapes into the atmosphere, it is a potential greenhouse gas.

Farmers can produce biogas from manure from their cattle by using anaerobic digesters

Factors affecting Biogas production:

1. Substrate temperature
2. pH level
3. Mixing Ratio
4. Loading Rate
5. Hydraulic Retention time
6. Nitrogen inhibition
7. C/N ratio
8. Agitation
9. Toxicity
10. Solid concentration
11. Seeding
12. Metal Cations
13. Particle size
14. Additives
15. BOD
16. COD
17. Heating

**BENEFITS OF BIOGAS TECHNOLOGY:**

Production of energy.

Transformation of organic wastes to very high quality fertilizer.

Improvement of hygienic conditions through reduction of pathogens.

Environmental advantages through protection of soil, water, air etc. Micro-economic benefits by energy and fertilizer substitutes. Macro-economic benefits through decentralizes energy generation and environmental protection.

Main types of simple biogas plants:

- **Balloon plants,**
- **Fixed-dome plants,**
- **Floating-drum plants types**

Balloon Plants:

A balloon plant consists of a plastic or rubber digester bag, in the upper part of which the gas is stored. The inlet and outlet are attached direct to the skin of the balloon. When the gas space is full, the plant works like a fixed-dome plant - i.e., the balloon is not inflated; it is not very elastic. The fermentation slurry is agitated slightly by the movement of the balloon skin. This is favorable to the digestion process. Even difficult feed materials, such as water hyacinths, can be used in a balloon plant. The balloon material must be UV-resistant. Materials which have been used successfully include RMP (red mud plastic), Trevira and butyl.

Advantages:

Low cost, ease of transportation, low construction (important if the water table is high), high digester temperatures, uncomplicated cleaning, emptying and maintenance.

Disadvantages:

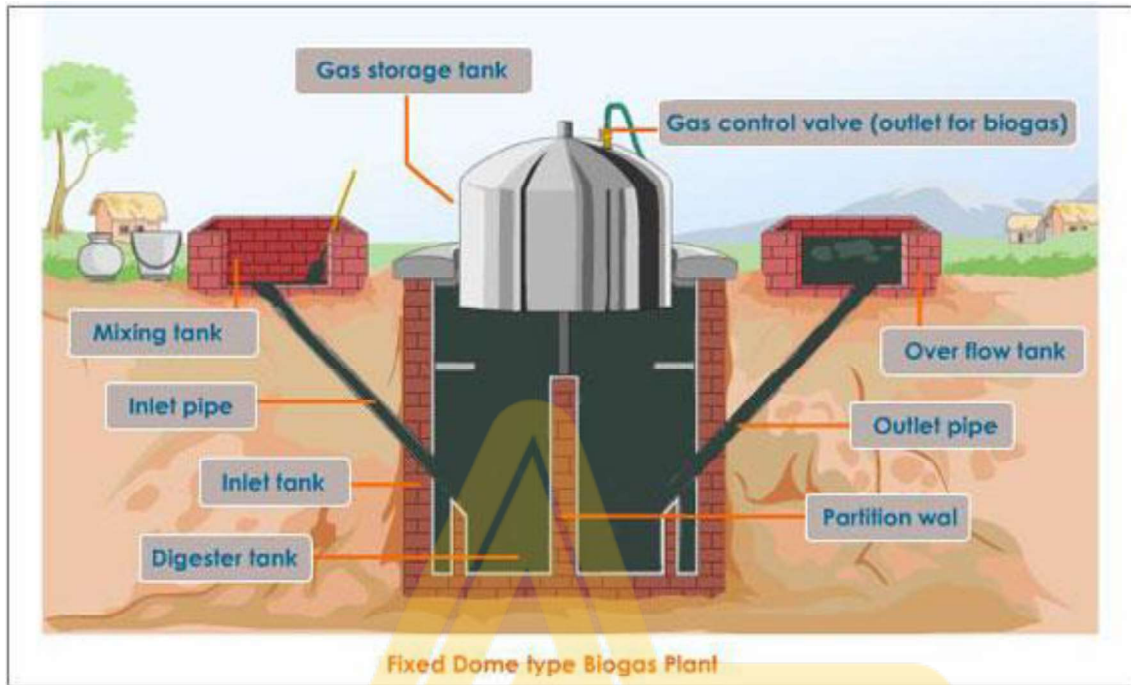
Short life (about five years), easily damaged, does not create employment locally, little scope for self-help.

Balloon plants can be recommended wherever the balloon skin is not likely to be damaged and where the temperature is even and high. One variant of the balloon plant is the channel-type digester with folia and sunshade.

Fixed-Dome Plants:

A fixed-dome plant consists of an enclosed digester with a fixed, non-movable gas space. The gas is stored in the upper part of the digester. When gas production commences, the slurry is displaced into

the compensating tank. Gas pressure increases with the volume of gas stored; therefore the volume of the digester should not exceed 20 m³. If there is little gas in the holder, the gas pressure is low.



Floating-Drum Plants:

Floating-drum plants consist of a digester and a moving gasholder. The gasholder floats either direct on the fermentation slurry or in a water jacket of its own. The gas collects in the gas drum, which thereby rises. If gas is drawn off, it falls again. The gas drum is prevented from tilting by a guide frame.

Advantages:

Simple, easily understood operation, constant gas pressure, volume of stored gas visible directly, few mistakes in construction.

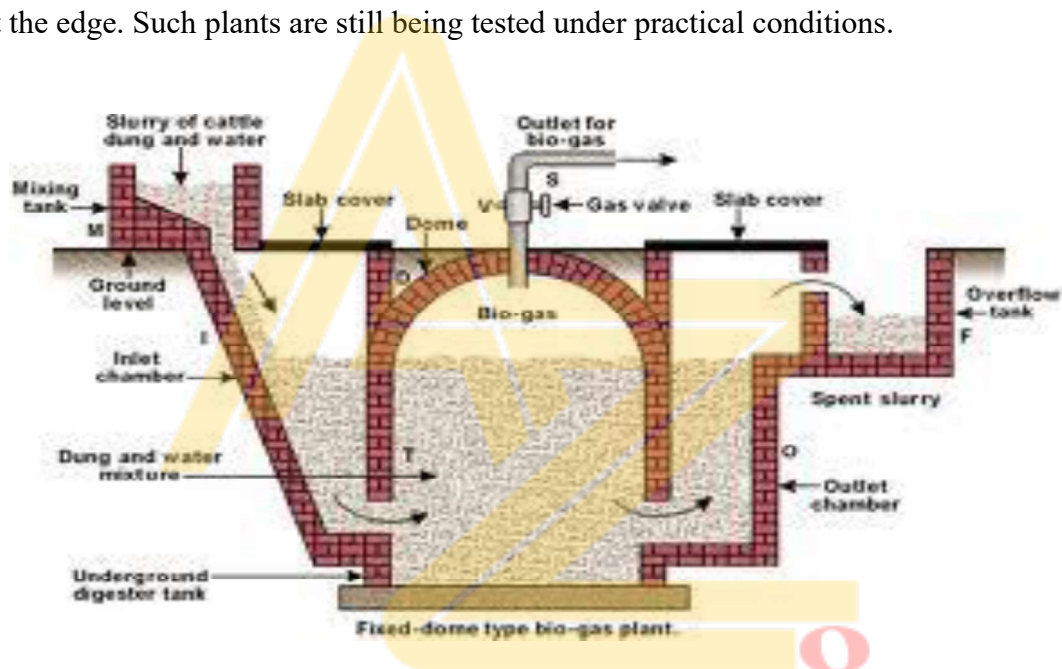
Disadvantages:

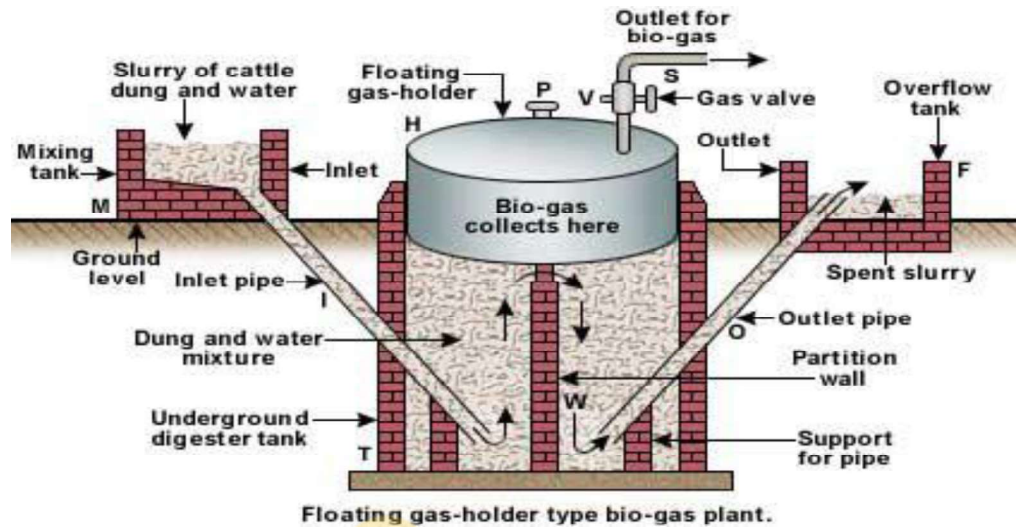
High construction cost of floating-drum, many steel parts liable to corrosion, resulting in short life (up to 15 years; in tropical coastal regions about five years for the drum), and regular maintenance costs due to painting.

In spite of these disadvantages, floating-drum plants are always to be recommended in cases of doubt. Water-jacket plants are universally applicable and especially easy to maintain. The drum won't stick, even if the substrate has high solids content.

Floating-drums made of glass-fiber reinforced plastic and high density polyethylene have been used successfully, but the construction cost is higher than with steel. Floating-drums made of Wire-mesh-reinforced concrete are liable to hairline cracking and are intrinsically porous. They require a gaslight, elastic internal coating. PVC drums are unsuitable because not resistant to UV.

The floating gas drum can be replaced by a balloon above the digester. This reduces construction costs (channel type digester with folia), but in practice problems always arise with the attachment of the balloon at the edge. Such plants are still being tested under practical conditions.





Energy from biomass:

1. Alternative source of energy
2. We have plenty of agricultural and forests for production of biomass.
3. Produced through photosynthesis achieved by solar energy conversion
4. Biomass means organic matter (Carbohydrate)
5. $H_2O + CO_2 \longrightarrow CH_2O + O_2$
6. $CH_2O + O_2 \longrightarrow CO_2 + H_2O + 112 \text{ Kcal/mole}$
7. Algae has lots of carbohydrates, could be harvested, dried and burned for production of heat
8. that could be converted into electricity.
9. Can be converted into liquid and gaseous fuels.

Advantage and applications:

- In big cities, sewage source is the main source for production of biogas.
- The sewage biogas is found to contain 84% methane, could be economically used to run engines to drive electric generator.
- In the rural sector, cooking and lighting mechanical power for generation of small electricity.
- The gas can be used with advantage to improve sanitary conditions and also to check environmental pollutions.
- 12 lakhs families in india are installed bio gas plants.
- Maradnagar (U.P.), Rishikesh (U.P.), Sanganer (Raj), Sihar (Raj) Pondicheri, bhopal etc.,

TIDAL ENERGY

INTRODUCTION

Tide is a periodic rise and fall of the water level of sea which are carried by the action of the sun and moon on the water of the earth.

ENERGY FROM THE OCEANS

Tide energy can furnish a significant portion of all such energies which are renewable in nature. It has been estimated that about a billion kW of tidal power is dissipated by friction and eddies alone. This is slightly less than the economically exploitable power potential of all the rivers of the World. It is only indication of the magnitude of tidal power available; all of it is not economically feasible also. The first attempt to utilize energy of the ocean was in the form of tidal “mills “in the eleventh century in Great Britain and latter in France and Spain.

The large scale up and down movement of sea water represents an unlimited source of energy. If some part of this vast energy can be converted into electrical energy it would be an important source of hydro-power. The main feature of the tidal cycle is the difference in water surface elevations at the high tide and at the low tide. If this differential head could be utilized in operating a hydraulic turbine, the tidal energy could be converted into electrical energy by means of an attached generator.

In principle, this is not very difficult as water, at the time of high tide, is at a high level and can be let into a basin to be stored at a high level. The same water can be let back into the sea during the low tide through the turbines, thus producing power.

Since the basin water level is high and sea water is low, there is a differential head comparable to the tidal range that can be utilized for the running of the turbines. Basically it appears to be a simple proposition, the problems involved in it, are many.

The Tides, as we see, although free, were inconvenient because they come at varying times from day to day, have varying ranges (heads) and, for large outputs required large capital expenditures. Their early use declined and eventually came to a half with the coming of the age of steam and cheap coal. With the beginning of the energy crisis in the 1970s, the tidal energy, like other renewable energy sources, received renewed attention.

The first tidal power plant was commissioned by General DeGaulle at La Rance in 1966 which marked a breakthrough. The average tidal range is 8.4 m (± 4.2 m), and the maximum is 13.5 m.

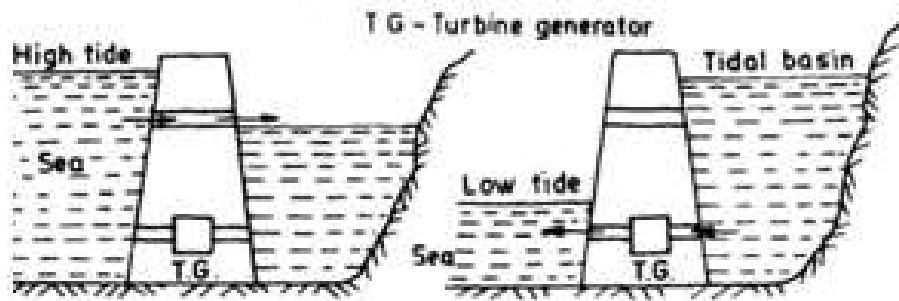


Fig. Principle of Tidal Power Generation.

Effective basin surface is 22 km²; basin volume is 184,000,000 m³. There are no special problems with this site, and it was a very sensible choice for the world's *first* tidal power station. It has used a single basin and submerged reversible propeller type turbine generators that could generate power with the water flowing in either direction through the turbine runner. Bulb turbines have been provided in this project, they are operating satisfactorily for the last 25 years.

The dam contains a small lock for commercial vessels, six movable sluice gates, and 24 turbo generators units of 10 MW each. The movable gates are used to accelerate filling and emptying of the storage basin at small differential water levels. Installed capacity of 240 MW, but of course the average power generating capacity is less, because power can be generated only intermittently.

Maximum electric energy production capacity is 544,000 MW h/yr., which gives a plant capacity factor, without any allowance for maintenance or down time, of 0.26. Maximum utilization of the stored hydraulic potential energy is 18 per cent, which may be increased to 24% in future by additional of 80 MW more generating capacity. There are presently two tidal power stations operating, one in France and one in the USSR. The one in France is a full commercial station. The one in the USSR is more in the nature of a pilot plant engineering experiment, and is rather small in size.

The French project at Rance near St. Malo in Brittany, uses a dam that goes straight across the estuary of the Rance river. The dam is not too long—750 m, shore to shore. The depth is never more than 12 m below mean sea level, there is an above surface rock part way across, the climate is moderate,

KisoloyaGuba, on the Barents sea, 70km north of Murmansk. It is small and was intended as a pilot project. It has one generator of 400 MW rating and delivers 700 to 800 MWh/yr, to the local electrical grid operation began in 1968.

Nearly for 17 years there was a dull period in the tidal power development due to the problems involved in it. The main disadvantage of tidal power plant is that the production of power from tidal station occurs at times and in magnitudes which are dependent on the relative positions of earth, moon and sun to one another and not on the electrical demand of consumers.

The success of La Rance stimulated interest in Canada where on the eastern coast, in the Bay of Fundy, the largest tidal ranges in the World occur—being as high as 16 m—on spring tides. The pioneering project in Nova Scotia, Canada at Annapolis Royal was commissioned in 1983. For the first time conventional bulb turbines were replaced by large rim generator units of straflo turbines with runner diameter of 7.6 m and capacity of 10 MW.

Though this type of turbine has been in use for many years in Europe, its runner diameter was limited to about 2 m by sealing problems at the runner periphery.

If these prove successful, it will provide an attractive alternative to the bulb unit, since the straflo unit has much higher inertia, thus enhancing system stability, and access to the generator is easier, thus facilitating maintenance.

In India, there are possible tidal projects in the Gulf of Kutch and Cambay, and on a smaller scale, in the Sunderbans regions of the Bay of Bengal.

In Korea, there have been a series of studies by Canadian and French firms, the latest one being of Garolim Bay. A comparative statistics of some recently proposed tidal schemes are given in the table.

Country Site	Installed capacity (MW)	Actual energy (GW h)	No. of 7.6 m diameter turbines	range (m)
UK Seven Estuary	7200	13000	230	9.3
	525	1090	21	6.7
Mersy Estuary	210	530	30	3.1
Strangford Lough Eire Shannon Estuary	318	715	30	3.8
India Gulf of Kutch: Kandla	600	1600	43	5.2
Korea Garolim Bay	480	1200	32	4.6
Brazil Bacanga	30	55	2	4.1
U.S.A. Knoik Arm	2220	5500	80	7.8
Canada Cumberland basin	1147	3420	37	10.5
Cobequid Bay		12600	106	12.4
Annapolis Royal	20	50	1	6.7
China Liangxia	3	11	6(2.5 m)	.- 6
USSR Lumbousky	400	—	—	-6
Mezenskaya	10000	—	—	— 9

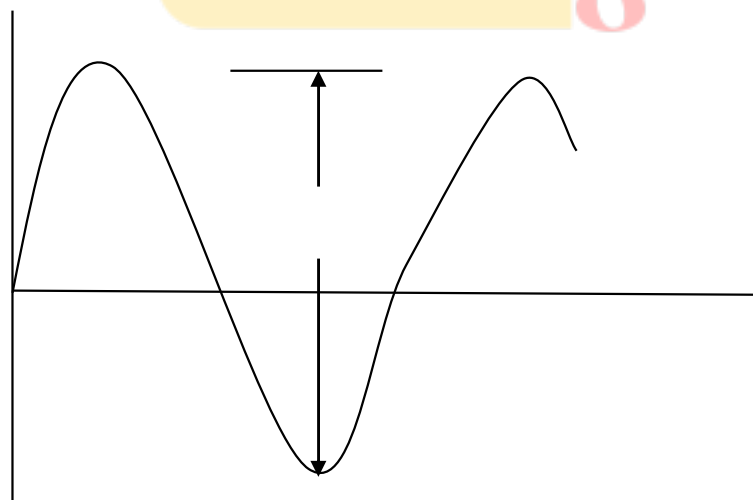
Table: Comparative Statistics of some Recently Proposed Schemes

BASIC PRINCIPAL OF TIDAL POWER

Tides are produced mainly by the gravitational attraction of the moon and the sun on the water of solid earth and the oceans. About 70 per cent of the tide producing force is due to the moon and 30 per cent to the sun. The moon is thus the major factor in the tide formation.

Surface water is pulled away from the earth on the side facing the moon, and at the same time the solid earth is pulled away from the water on the opposite side. Thus high tides occur in these two areas with low tides at intermediate points. As the earth rotates, the position of a given area relative to the moon changes, and so also do the tides. There is thus a periodic succession of high and low tides. Although there are exceptions, two tidal cycles (*i.e.* two high tides and two low tides) occur during a lunar day of 24 hours and 50 minutes (The lunar day is the apparent time of revolution of the moon about the earth).

That is to say, the time between high tides and low tide at any given location is a little over 6 hours. A high tide will be experienced at a point which is directly under the moon. At the same time, a diametrically opposite point on the earth's surface also experiences a high tide due to dynamic balancing. Thus a full moon as well as a no moon produces a high tide. In a period of 24 hrs 50 minutes, there are therefore, two high tides and two low tides. Fig. These are called semi-diurnal tides. The rise and fall of the water level follows a sinusoidal curve, shown with point A indicating the high tide point and point B indicating the low tide point. The average time for the water level to fall from A to B and then rise to C is approximately 6 hours 12.5 mm.



The difference between high and low water levels is called the *range* of the tide. The *tidal range* R is defined as: $R = \text{water elevation at high tide} - \text{water elevation at low tide}$.

Because of the changing positions of the moon and sun relative to the earth, the range varies continuously. There are however, some characteristic features of this variation. At times near full or new moon, when sun, moon and earth are approximately in a line, the gravitational forces of sun and moon enhance each other. The tidal range is then exceptionally large, the high tides are higher and low tides are lower than the average. These high tides are called *spring tides*, on the other hand, near the first and third quarters of the moon, when the sun and moon are at right angles with respect to the earth, *neap tides* occur. The tidal range is then exceptionally small; the high tides are lower and the low tides *higher* than the average. Hence the range is not constant.

It varies during the 29.5 day lunar month (Fig.) being maximum at the time of new and full moons, called the spring tides, and minimum at the time of the first and third quarter moons, called the neap tides. The spring-neap tidal cycle lasts one-half of a lunar month. A typical *mean range* is roughly one third of the spring range. The actual variations in range are somewhat complicated by seasonal variations caused by the ellipticity of the earth's orbit around the sun.

The variations in the periodicity and monthly and seasonal ranges must, of course, be taken into account in the design and operation of tidal power plants. The tides, however, are usually predictable, and fairly accurate tide tables are usually available.

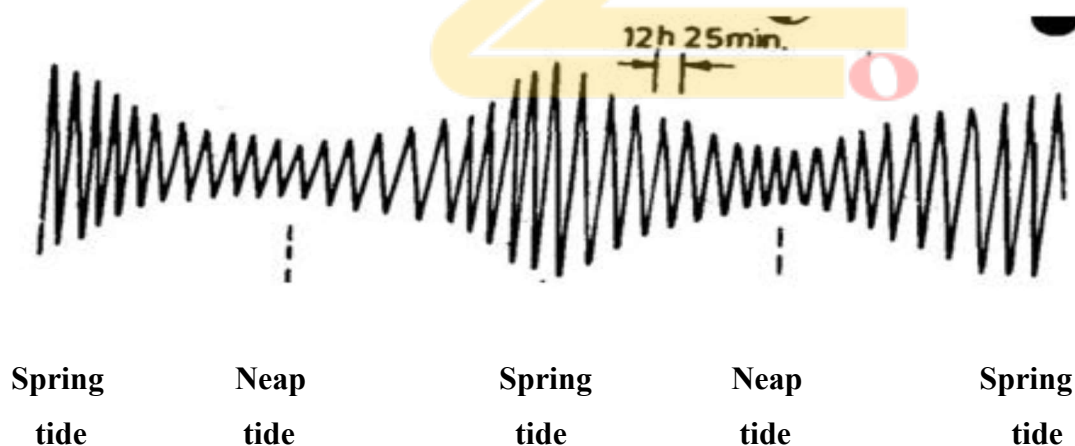


Fig. Relative high and low tides showing variation in range during lunar month.

Tidal ranges vary from one earth location to another. They are influenced by such conditions as the profile of the local shoreline and water depth. When these are favourable, a resonance like effect

causes very large tidal ranges. Ranges have to be very large to justify the huge costs of building dams and associated hydroelectric power plants. Such tides occur only in a few locations in the world.

Following *points* have to be specially noted in connection with the tidal phenomenon.

(1) The tides are a periodical phenomenon but no two tides in any cycle are alike. Since the relative positions of sun and moon and their distances from earth are continuously changing, the tides are also influenced accordingly.

Of the two high tides in a single day, one tide is higher than the other. In any month, the tides on the full moon and no moon days are particularly higher than the rest, as on these days sun's and moon's attraction acts in a directly additive manner. These are termed as the spring tides.

In any year, the tides that occur at the time of vernal and autumnal equinoxes will be even higher due to the relative location of the sun and earth. Thus the tidal range R shown in the figure varies from time to time. Generally, a long time mean value of R is designated as mean tidal range at any particular place.

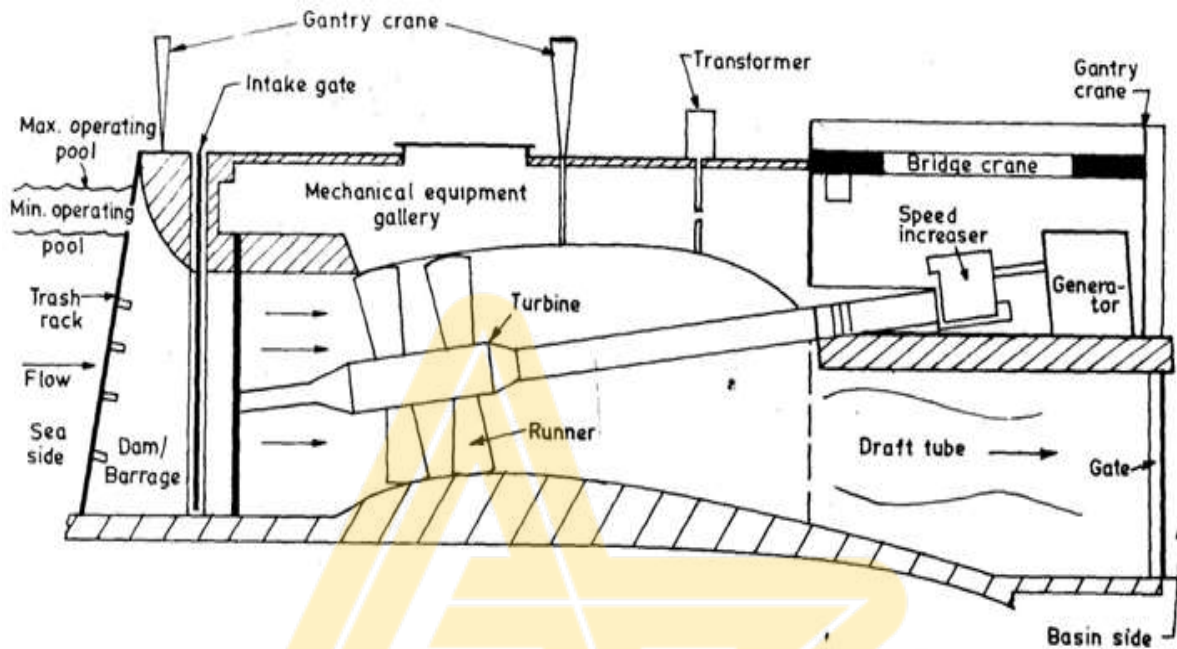
(2) The mean tidal range varies from place to place. The shape of the tidal cycle depends upon the interaction of the sea with the coast-line. Where the coast-line offers a resonating influence, the tidal range gets accentuated, at the other places, the land may produce a dampening effect on the tidal phenomenon. For instance, in land locked seas, the tidal phenomenon is always much subdued. Because of this interacting effect, the tidal range (as well as the tidal period) varies from place to place. For example; the mean tidal range on the west coast of India is as high as 7 to 8 meters near the gulf of Kutch, whereas it is only one meter or so near Kerala, down south. Bay of Fundy (Canada) has one of the greatest tidal ranges in the world *i.e.* of 20 m, whereas the Adriatic sea at the Zara has virtually static water with the range being only of a few cm. Thus the tidal phenomenon is a unique feature of every coast line.

(3) In spite of their complexity, the tides are amenable to mathematical analysis. As a result the exact time and the water level for a high tide as low tide can be forecast with great accuracy.

COMPONENTS OF TIDAL POWER PLANTS

There are *three* main components of a tidal power plant *i.e.*

- (i) The power house
- (ii) The dam or barrage (low wall) to form pool or basin.
- (iii) Sluice-ways from the basins to the sea and *vice versa*



The turbine, electric generators and other auxiliary equipments are the main equipments of a power house. The function of dam to form a barrier between the sea and the basin or between one basin and the other in case of multiple basins.

The sluice ways are used either to fill the basin during the high tide or empty the basin during the low tide, as per operational requirement. These are gate controlled devices.

It is generally convenient to have the power house as well as the sluice-ways in alignment with the dam. The design cycle may also provide for pumping between the basin and the sea in either direction. If reversible pump turbines are provided, the pumping operation can be taken over at any time by the same machine.

The modern tubular turbines are so versatile that they can be used either as turbines or as pumps in either direction of flow. In addition, the tubular passages can also be used as sluice-ways by locking the machine to a standstill.

As compared to conventional plants, this, however, imposes a great number of operations in tidal power plants. For instance, the periodic opening and closing of the sluice-ways of a tidal plant are about 730 times in a year.

Dam (Barrage). Dam and barrage are synonymous terms. Barrage has been suggested as a more accurate term for tidal power schemes, because it has only to withstand heads a fraction of the structure's height, and stability problems are far more modest.

However, the literature does not always make the distinction, even though heads are small with tidal power cut offs.

Tidal power barrages have to resist waves whose shock can be severe and where pressure changes sides continuously.

Since barrage length adds also to the price tag of the plant, short barrages are preferred even if basin size may have to be smaller as a result of site choice. Up to a height of 20 m, cost remains proportional to length as it is not changed by the need to build a darn wall to withstand high hydrostatic pressure.

When, the elevation exceeds the 20 m limit, costs increase faster with length. Most tidal power plants do not have heads exceeding 20 m.

The barrage needs to provide channels for the turbines in prestressed or reinforced concrete. To build these channels a temporary cofferdam is necessary, but it is now possible to build them on land, float them to the site, and sink them into place.

Flatness is required for the sea bottom; sandy bottom usually necessitates piling. There, where sand or rock can bear the weight of the structure to be implanted, the bottom can be prepared, the structure placed on it and then anchored. Prefabricated concrete blocks can be used as the core for large barrages and voids filled with rocks or concrete remaining holes with sand, and the entire construction then asphalted.

Construction of a barrage usually will influence the tidal amplitude. Indeed, such a construction modifies the effective length of the embayment or basin and its shape as well, particularly if the scheme involves supplementary spur dams, or brings about relocation or disappearance of natural obstruction as is foreseen for the seven plants and has occurred in the Rance estuary.

The construction influence the resonance of the bay, and most bays are less than the resonant length of the tidal wave. If resonance is reduced, the range will decrease; if measures are taken to augment the resonance, tidal amplitude may be increased.

Tidal barrages require sites where there is a sufficiently high tidal range to give a good head of water—the minimum useful range is around three meters. The best sites are bays and estuaries, but water can also be impounded behind bunded reservoirs built between two points on the same shore line.

The precise design of barrage and its mode of operation depend critically on the requirements for power and on a careful analysis of the economics. The simplest and cheapest schemes would normally involve—a single barrage designed to trap water in a basin at high tide and to generate on the ebb. More complex schemes could involve generation on both the ebb and flood tides, or the construction of a secondary basin which would permit water to be stored and discharged whenever desired. This would provide more firm power on a flexible basis.

The expertise necessary to design and build such structures is available. The location of the barrage is important, because the energy available is related to the size of the trapped basin and to the *square* of the tidal range. The nearer it is built to the mouth of an estuary or bay, the larger the basin, but the smaller the tidal range. A balance must also be struck between increased output and increased material requirements and construction costs.

Gates and Locks.

Tidal power basins have to be filled and emptied. Gates are opened regularly and frequently but heads vary in height and on the side where they occur, which is not the case with conventional river projects. The gates must be opened and closed rapidly and this operation should use a minimum of power. Leakage is tolerable for gates and barrage. Since we are dealing with sea water, corrosion problems are acute, they have been very successfully solved by the cathodic protection and where not possible by paint. Gate structures can be floated as modular units into place.

Though, in existing plants, vertical lift gates have been used, technology is about ready to substitute a series of flaps that operate by water pressure. Flap gates are gates that are positioned so as to allow water into the holding basin and require no mechanical means of operation. If used they are positioned, in the case of modular construction, in the caissons.

A caisson is then floated into place. They have to be built so that they will be adequate for the maximum tidal amplitude. Top hung on a gate-hoisting beam, a gate would transfer its hydraulic load to the concrete structure. If operation is to be rapid and efficient, gates must open under the maximum differential head; this favours as flat as possible the tidal basin face of the sluice.

The flap gates allow only in the direction of the sea to basin. Hence, the basin level rises well above to sea level as ebb flow area is far less than flood flow area.

Power House. Because small heads only are available, large size turbines are needed; hence, the power house is also a large structure. Both the French and Soviet operating plants use the bulb type of turbine. Of the propeller type, with reversible blades, bulbs have horizontal shafts coupled to a single generator. The cost per installed kilowatt drops with turbine size, and perhaps larger turbines might be installed in a future major tidal power plant. The Bulb Group (Rance Example). A bulb type turbine is an axial flow turbine the bulb set, resembling in appearance a small submarine, is made up of an ogive shaped steel shell containing an alternator and a Kaplan turbine. It is placed in a horizontal hydraulic duct and entirely surrounded by water; a shaft provides communication with the engine room of the power plant.

The set function as a turbine and as a pump, and regulates the flow in both directions of flow, tide to reservoir (basin) and reservoir to tide. The alternator is directly coupled to the turbine and turns it at some pressure. The turbine is a Kaplan wheel with four mobile blades and guide vanes. The group, functioning as a turbine in the direct sense basin to sea, and furnishing power in reverse, sea to basin direction, functioning as a pump in direct sense. (Fig.)

Rim Type Turbines. Different types of turbines are under study; usually mentioned are included shaft turbines, rim type turbines, or straight flow turbines, where the generator is attached peripherally on the turbine blades, an arrangement that couples two turbines of conventional type to one generator, and a hydraulic system in which upto six turbines are coupled to hydrostatic pumps and to drive a pelton-wheel, which, in turn, drives a high speed generator.

The main problem in rim type turbine, in which the rotor surrounds the turbine runner as a rim carried by the runner blades, is the seals between the stationary parts and the rotating rim. Engineers who favour straight flow generators against the bulb turbine generators point out the lower inertia characteristic of the bulb type, claiming this could lead to problems during power system disturbances.

The designers of a new type of straight flow units put forth savings in civil works and in generator and auxiliary electrical equipment because large unit capacities than with the bulb type unit, for the same head, would be possible.

OPERATION METHODS OF UTILIZATION OF TIDAL ENERGY

The generation of electricity from water power requires that there should be a difference in levels (or heads) between which water flows. A number of concepts have been proposed for generating electricity by utilizing the head that can be produced by the rise and fall of the tides to operate a hydraulic turbine.

The power generation from tides involves flow between an artificially developed basin and the sea. However in order to have a more or less continuous generation, this basic scheme can be elaborated by having two or more basins.

Accordingly we can distinguish the following types of arrangements:

- (1) Single basin arrangement,
- (2) Double basin arrangement.

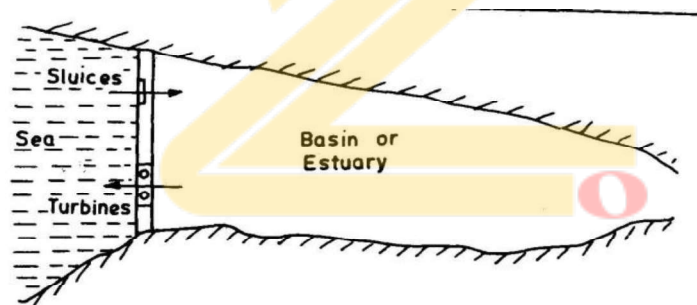
Single-basin schemes can generate power only intermittently, but a double basin scheme can provide power continuously, or - on demand, which is a tremendous advantage. The drawback is that the civil works become more extensive. In the simplest double-basin scheme there must be a dam between each basin and the sea, and also a dam between the basins, containing the power house. One basin is main tame always at a lower level than the other.

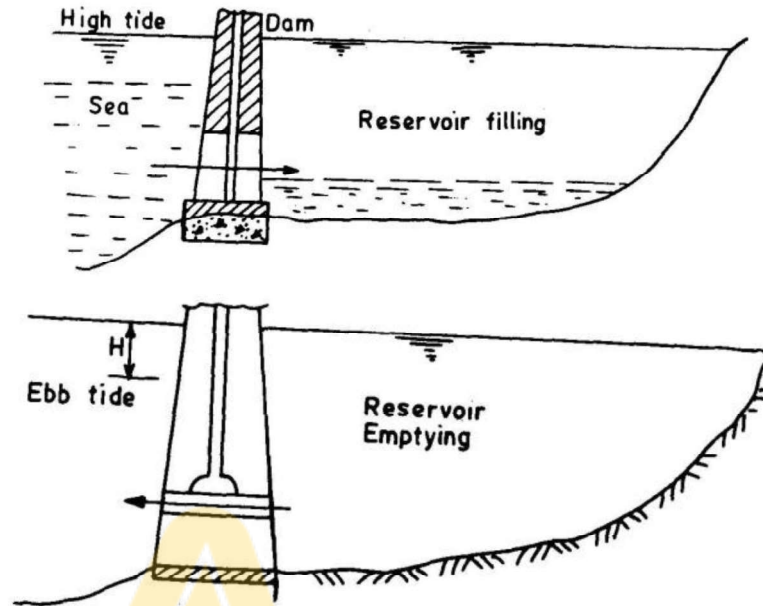
The lower reservoir empties at low tide, and the upper reservoir is replenished at high tide. If the generating capacity is to be large, the reservoirs must be large, which usually means that the dams will be long. (1) Single Basin Arrangement The simplest way to generate tidal power is to use a single basin with a retaining dam in the following manner: In a single basin arrangement there is only one basin interacting with the sea. The two are separated by a dam (or barrage) and the flow between them is through sluice ways located conveniently along the dam. Potential head is provided by rise and fall of tidal water levels, this is usually accomplished by blocking the mouth of a long narrow estuary with a dam across it, thereby creating a reservoir. The darn or barrage embodies a number of sluice

gates and low head turbine sets. The generation of power can be achieved in a single basin arrangement either as a

- (a) Single ebb-cycle system, or
- (b) Single tide-cycle system, or
- (c) Double cycle system.

(a) SINGLE EBB CYCLE SYSTEM. When the flood tide (high tide) comes in, the sluice gates are opened to permit sea-water to enter the basin or reservoir, while the turbine sets are shut. The reservoir thus starts filling while its level rises, till the maximum tide level is reached. At the beginning of the ebb tide the sluice gates are closed. Then the generation of power takes place when the sea is ebbing (flowing back of tide) and the water from the basin flows over the turbines into the lower level sea water. After two or three hours when there is sufficient difference between the full reservoir level and the falling tide level, to run the turbines, they are started and keep working until the rising level of the next flood tide and the falling reservoir level together reduce the effective head on the turbines to the extent where it can no longer work safely and efficiently. The turbines are then closed and the sluice gates opened again; to repeat the cycle of operations. Since in an estuary, the ebb tide has a long duration than the flood tide, the ebb operation provides an increased period of actual work.





(b) Sluice gates open, Turbine shut off (Section).

(c) Sluice gates shut, Turbine operation (Section)

FIG. TIDAL POWER PLANT (SINGLE BASIN OPERATION).

(B) SINGLE TIDE CYCLE SYSTEM. In single tide cycle system, the generation is affected when the sea is at flood tide. The water of the sea is admitted into the basin over the turbines. As the flood tide period is over and the sea level starts falling again, the generation is stopped. The basin is drained into the sea through the sluice ways.

Flood operation scheme needs larger size plant, operating, for shorter period and hence less efficient as compared to ebb tide operation. The ebb operation plant will be of smaller size, but will operate over a large period. The aim and effort should be to obtain as long a period of operation as possible at the beginning and finishing the work at the minimum operating head. Adjustable blade bulb turbines are especially suited for such operations under low and variable heads. All these turbines may be reversible which can work as pumps after the high tide, to increase the amount of water in the storage lake, until the generation can start. This way the turbine operation period may be prolonged to ten or more hours, out of the twelve hours-twenty five minutes of the tidal cycle.

The tide-cycle requires a deeper reservoir so as to locate the sills of the sluice gates deeper and, thus, requires greater construction costs. It has been estimated that the energy produced by an ebb cycle system can be as much as 1.5 times that by a tide cycle system. The main disadvantage in both the ebb-cycle as well as the tide cycle systems is the intermittent nature of their operation. However, since the intermissions occur at regular intervals, there is possibility of connecting another supplementary system, so as to balance the discontinuity. Such possibilities can regulate the output. The system can be so geared as to generate power, both during the ebb and flood tides with the help of single basin only. This system is known as the double-cycle system.

(C) DOUBLE CYCLE SYSTEM. As stated above in double cycle system, the power generation is affected during the ebb as well as in flood tides. The direction of flow through the turbines during the ebb and flood tides alternates, but the machine acts as a turbine for either direction of flow. In this method, the generation of power is accomplished both during emptying and filling cycles. Both filling and emptying processes take place during short periods of time, the filling when the ocean is at high tide while the water in the basin is at low tide level, the emptying when the ocean is at low tide and the basin at high-tide level. The flow of water in both directions is used to drive a number of reversible water turbines, each driving an electrical generator. Electric power would thus be generated during two short periods during each tidal period of 12 h, 25 mm. or once every 6h, 12.5 mm.

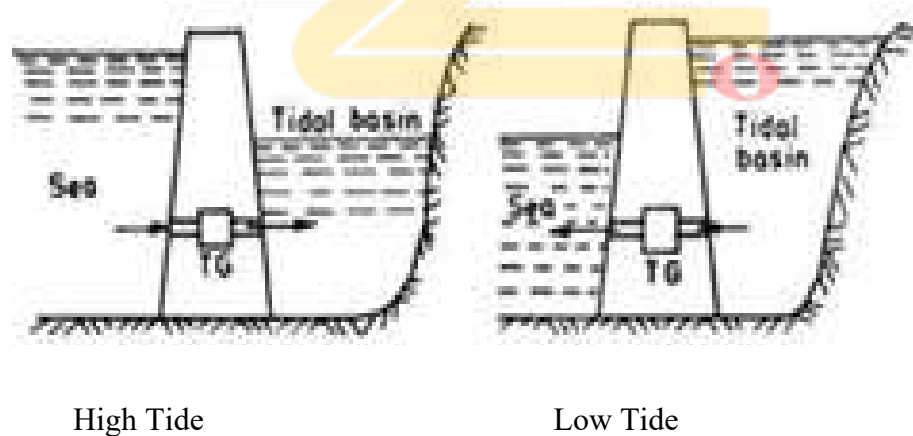


Fig. Double Cycle System

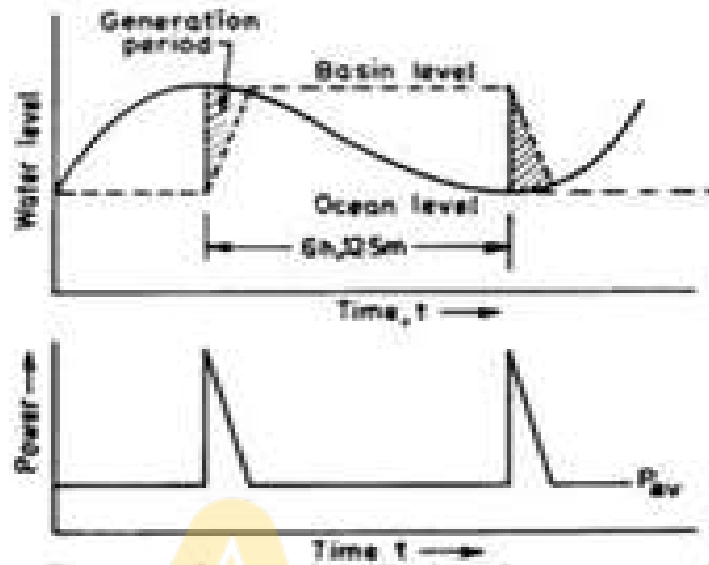


Fig. Ocean and pool levels and power generated in a single-basin tidal system.

Though the double cycle system has only short duration interruptions in the turbine operation, yet a continuous generation of power is still not possible. Furthermore the periods of power generation coincide only occasionally with periods of peak demand. These problems are solved to some extent in the two-basin scheme described below. However, a fundamental drawback to all methods for generating tidal power is the variability in output caused by the variations in the tidal range.

(2) Double Basin Arrangement It requires two separate but adjacent basins. In one basin called “upper basin” (or high pool), the water level is maintained above that in the other, the low basin (or low pool). Because there is always a head between upper and lower basins, electricity can be generated continuously, although at a variable rate.

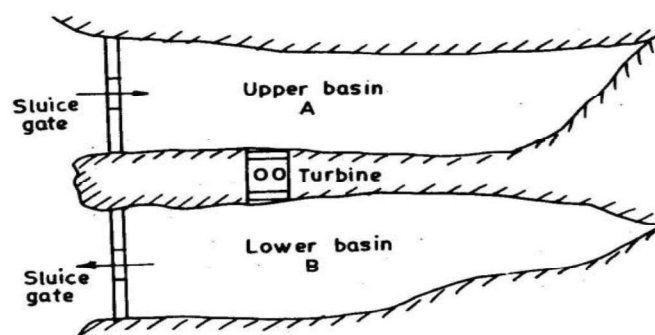


Fig. Tidal power plant Double Basin Operation.

In this system the turbines are located in between the two adjacent basins, while the sluice gates are as usual embodied in the dam across the mouths of the two estuaries. At the beginning of the flood tide, the turbines are shut down, the gates of upper basin A are opened and those of the lower basin B are closed. The basin A is thus filled up while the basin B remains empty. As soon as the rising water level in A provides sufficient difference of head between the two basins, the turbines are started. The water flows from A to B through the turbines, generating power. The power generation thus continues simultaneously with the filling up the basin A. At the end of the flood tide when A is full and the water level in it is the maximum, its sluice gates are closed. When the ebb tide level gets lower than the water level in B, its sluice gates are opened whereby the water level in B, which was arising and reducing the operating head, starts falling with the ebb. This continues until the head and water level in A is sufficient to run the turbines. With the next flood tide the cycle repeats itself. With this twin basin system, a longer and more continuous period of generation per day is possible. The small gaps in the operation of such stations can be filled up by thermal power.

The operation of the two basin scheme can be controlled so that there is a continuous water flow from upper to lower basin. However since the water head between the basins varies during each tidal cycle, as well as from day to day, so also does the power generated. As in the case with single basin scheme, the peak power generation does not often correspond in time with the peak demand. One way of improving the situation is to use off-peak power, from the tidal power generators or from an alternative system, to pump water from the low basin to the high basin. An increased head would then be available for tidal power generation at times of peak demand. This is very similar to pumped storage system in hydro-electric power stations.

Estimate of Energy and Power in Simple Single Basin Tidal System

The expression of maximum energy that can be generated during one generation period can be derived with the help of Fig. which shows the case of the basin beginning at high-tide level, emptying through the turbine to the ocean, which is at low tide. (The identical energy will be generated in reverse process). During the emptying process, the differential work done by the water is equal to its potential energy at the time. Considering a tidal range R , and intermediate head, at a given time, the amount of work is calculated, considering a small head dh , for a intermediate head h , as shown in figure.